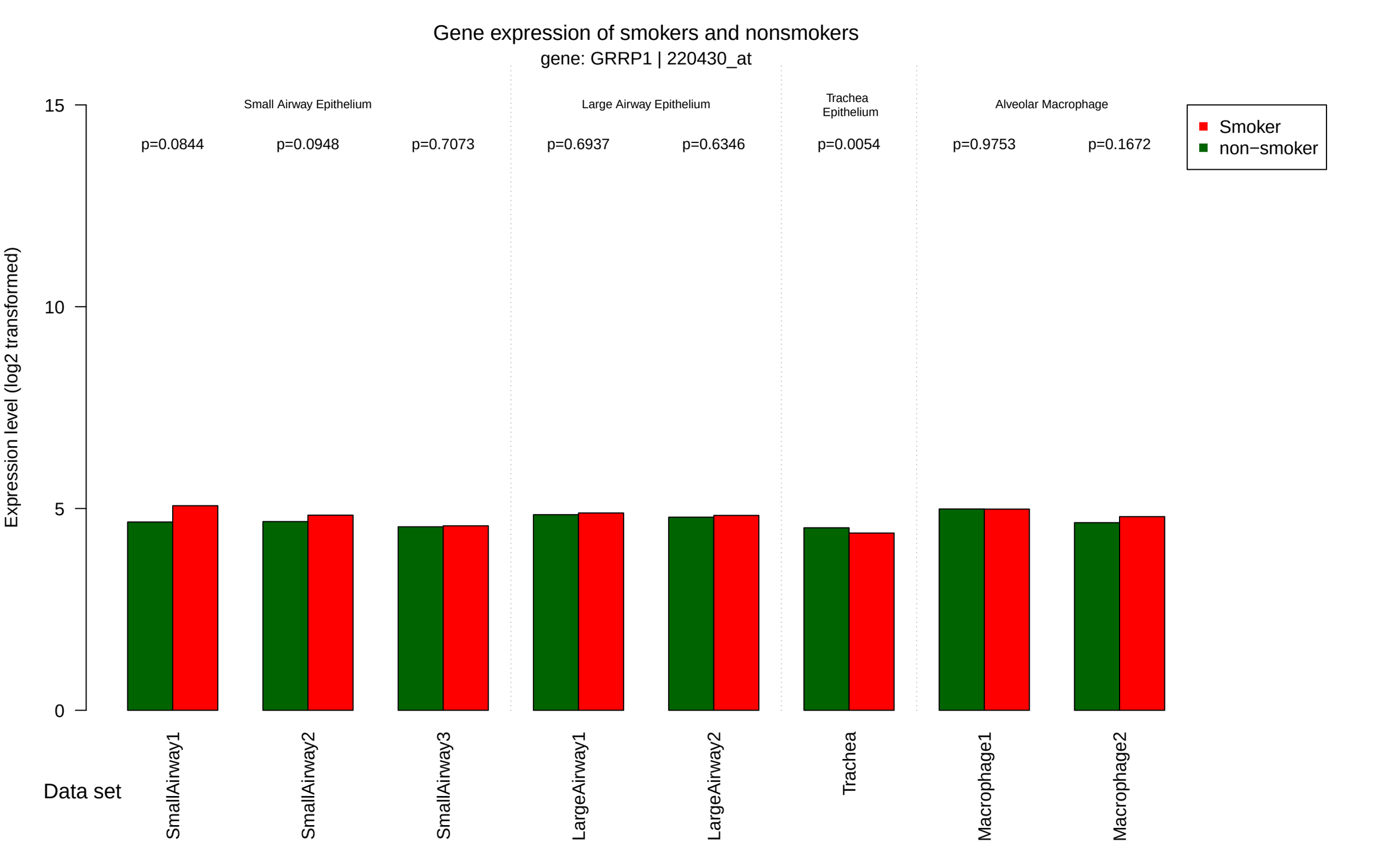
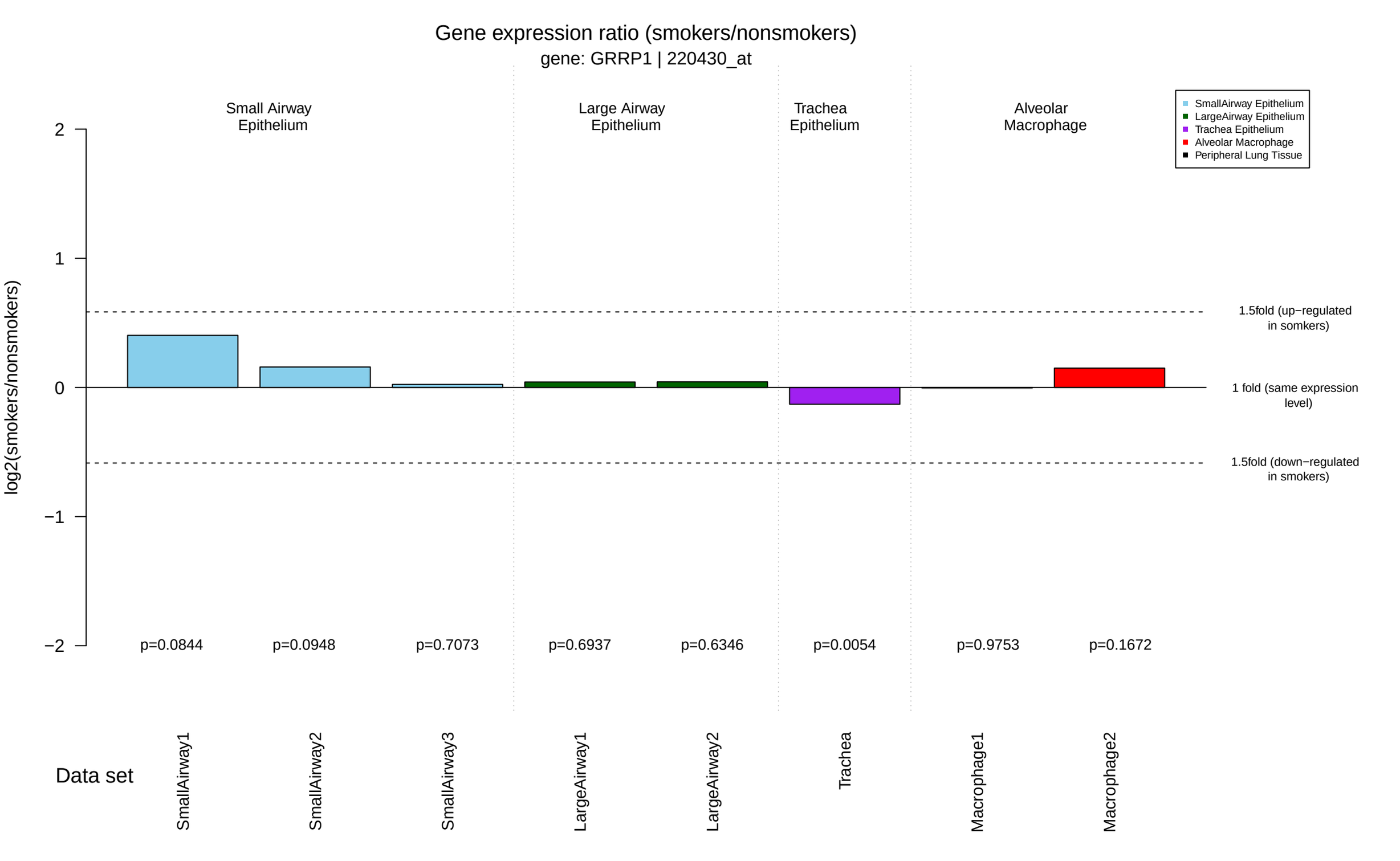
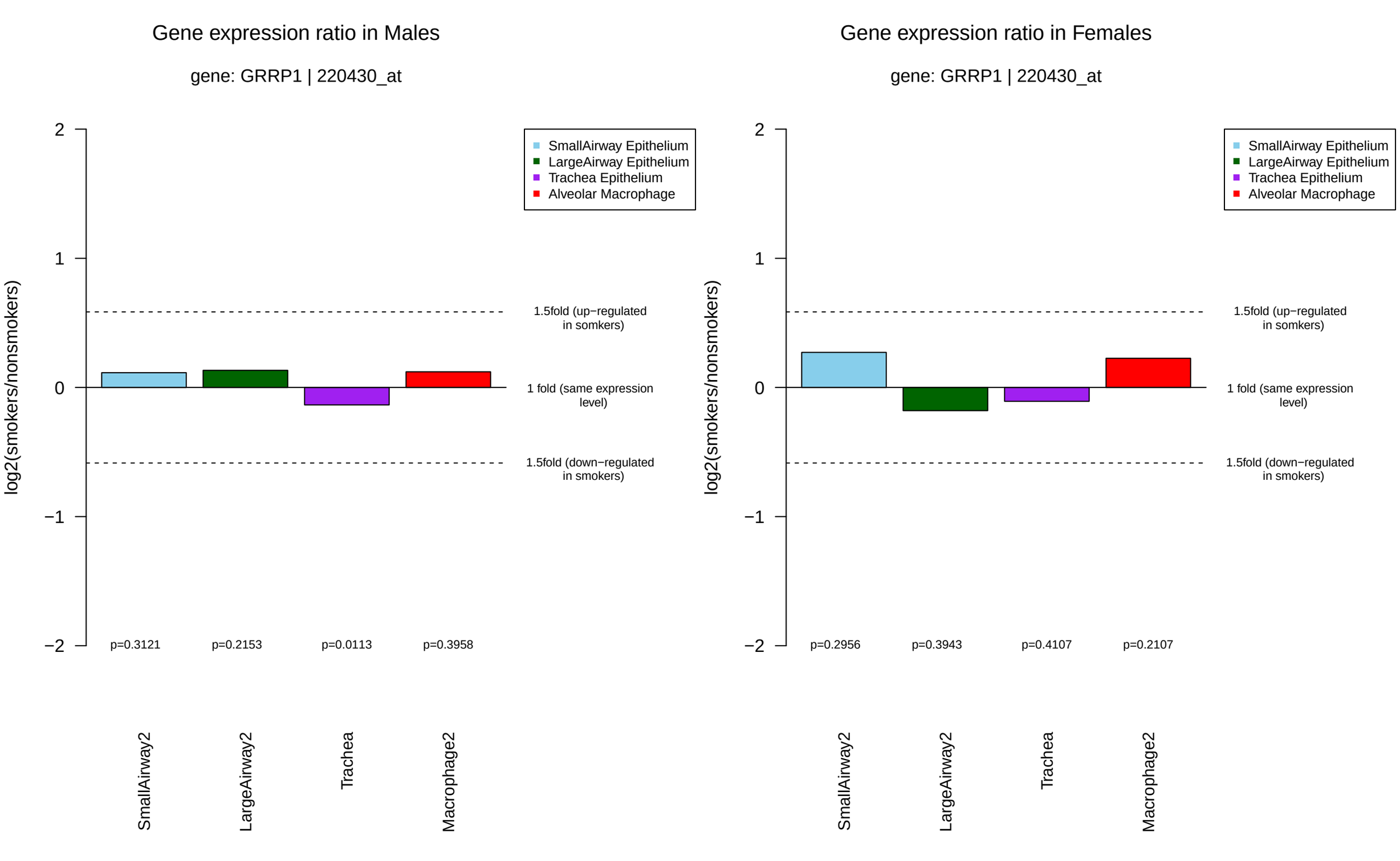
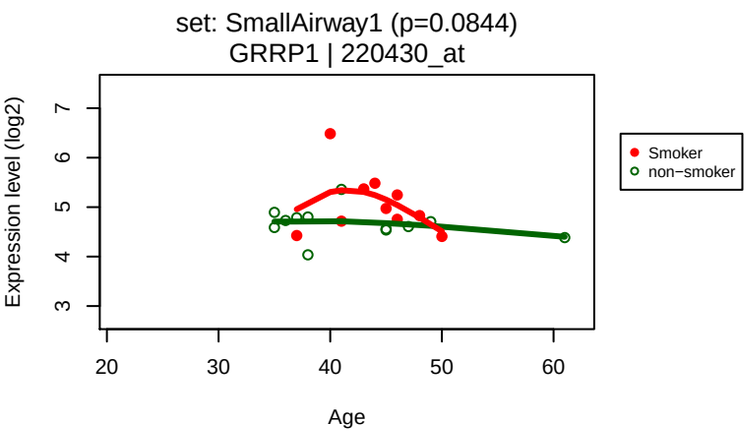
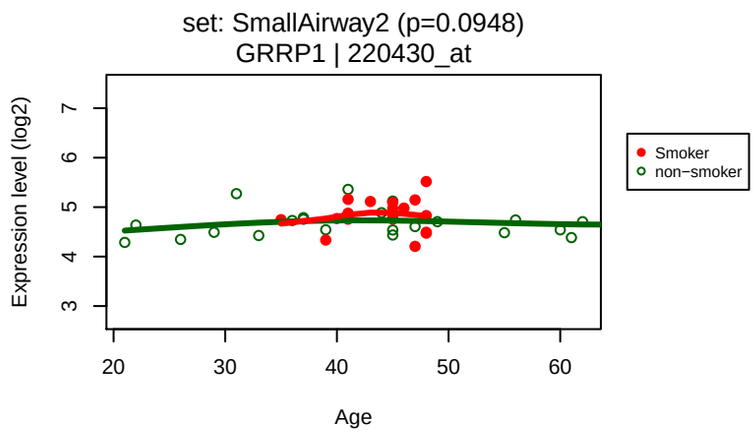
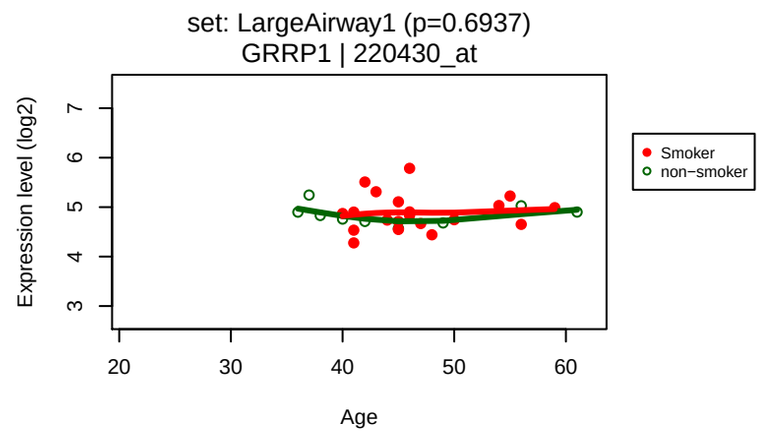

Gene expression of smokers and nonsmokers
gene: GRRP1 | 220430_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0844
p=0.0948
p=0.7073
p=0.6937
p=0.6346
p=0.0054
p=0.9753
p=0.1672
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

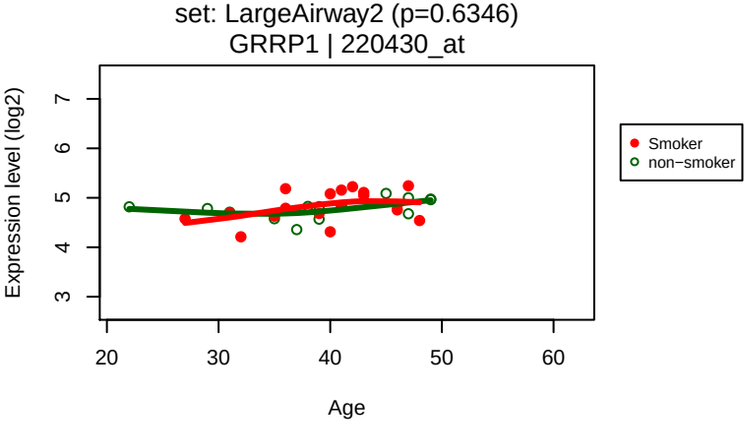

Gene expression ratio (smokers/nonsmokers)
gene: GRRP1 | 220430_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0844
p=0.0948
p=0.7073
p=0.6937
p=0.6346
p=0.0054
p=0.9753
p=0.1672
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

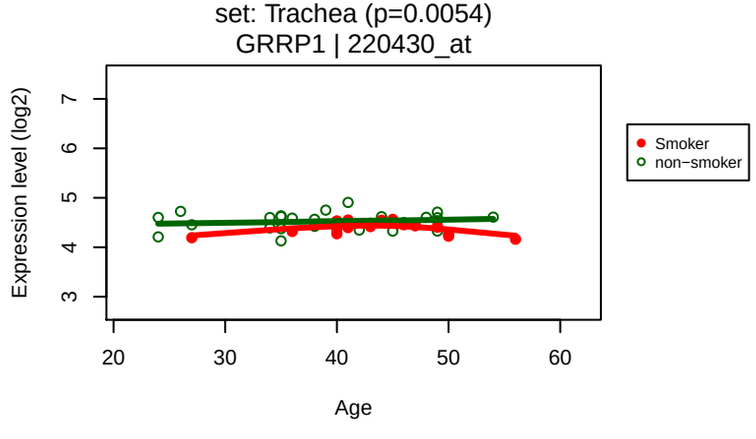

Gene expression ratio in Males
gene: GRRP1 | 220430_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3121
p=0.2153
p=0.0113
p=0.3958
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRRP1 | 220430_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2956
p=0.3943
p=0.4107
p=0.2107
SmallAirway2
LargeAirway2
Trachea
Macrophage2

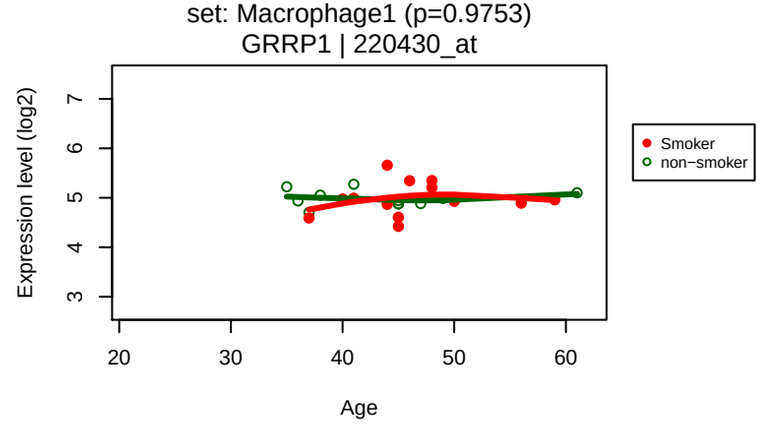

set: SmallAirway1 (p=0.0844)
GRRP1 | 220430_at
Expression level (log2)
Age
Smoker
non−smoker

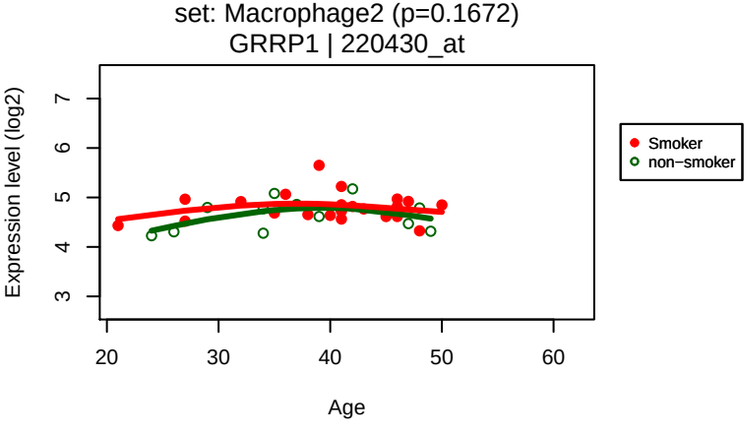

set: SmallAirway2 (p=0.0948)
GRRP1 | 220430_at
Expression level (log2)
Age
Smoker
non−smoker

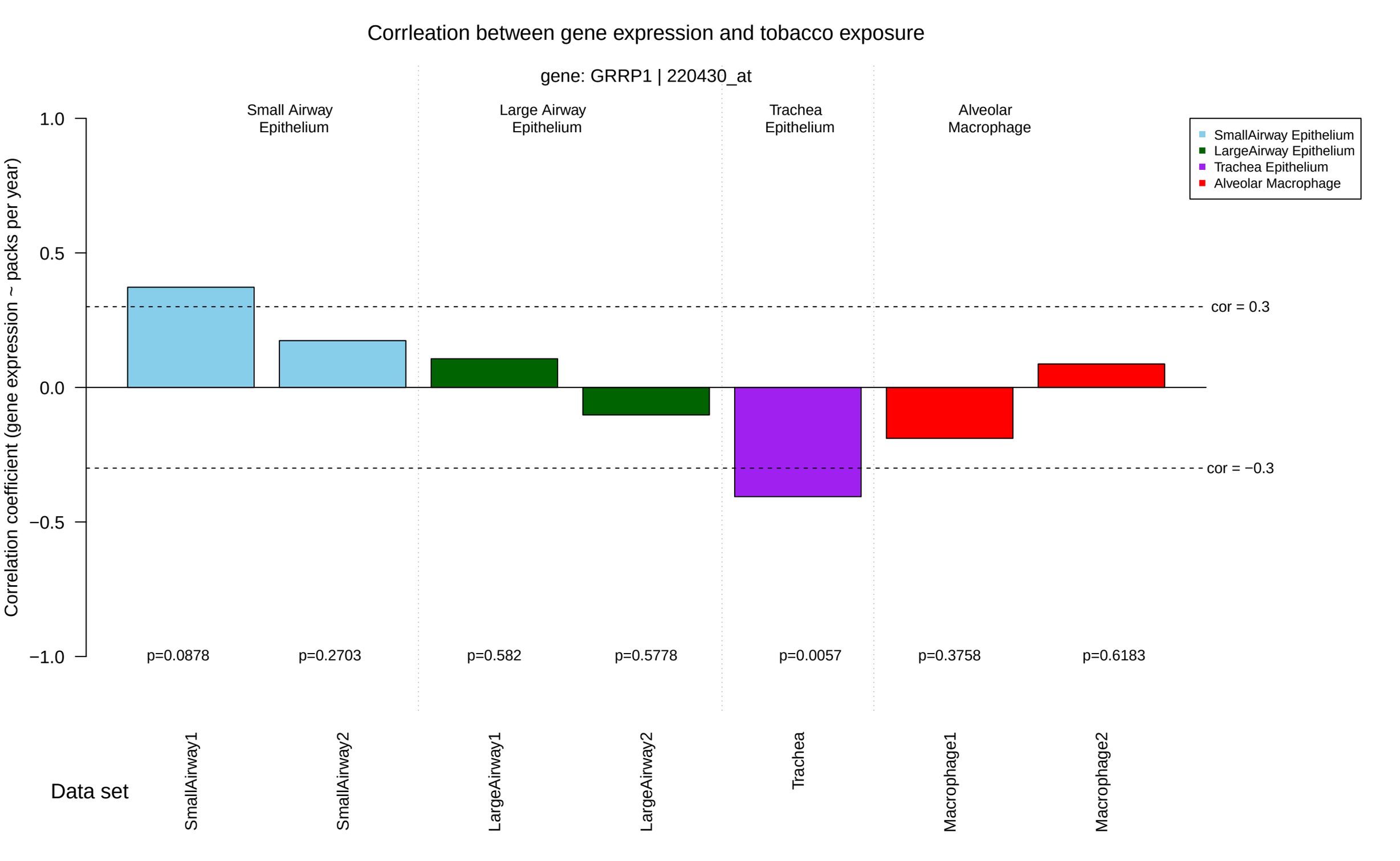

set: LargeAirway1 (p=0.6937)
GRRP1 | 220430_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6346)
GRRP1 | 220430_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0054)
GRRP1 | 220430_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9753)
GRRP1 | 220430_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1672)
GRRP1 | 220430_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRRP1 | 220430_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0878
p=0.2703
p=0.582
p=0.5778
p=0.0057
p=0.3758
p=0.6183
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2